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United States Patent	12397209
Kind Code	B2
Date of Patent	August 26, 2025
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Golf club heads with optimized characteristics and related methods

Abstract

Embodiments of golf club heads comprising a head interior and a hosel structure, the head interior bounded by a head front portion, a head rear portion, a head heel portion, a head toe portion, a head top portion, and a head sole portion, the golf club heads further comprising various optimized characteristics, are presented herein. Other examples and related methods are also disclosed herein.

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Appl. No.:	18/475134
Filed:	September 26, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240009533 A1	Jan. 11, 2024

Related U.S. Application Data

continuation parent-doc US 17661903 20220503 US 11819746 child-doc US 18475134
continuation parent-doc US 17094707 20201110 US 11318359 20220503 child-doc US 17661903
continuation parent-doc US 16503322 20190703 US 10857435 20201208 child-doc US 17094707
continuation parent-doc US 15680414 20170818 US 10357700 20190723 child-doc US 16503322
continuation parent-doc US 14836732 20150826 US 9764205 20170919 child-doc US 15680414
continuation parent-doc US 13804917 20130314 US 9168429 20151027 child-doc US 14836732

Publication Classification

Int. Cl.: A63B53/04 (20150101); A63B60/00 (20150101); A63B60/02 (20150101)

U.S. Cl.:

CPC A63B60/00 (20151001); A63B53/04 (20130101); A63B53/0466 (20130101); A63B53/0408 (20200801); A63B53/0412 (20200801); A63B53/0416 (20200801); A63B53/0433 (20200801); A63B53/0462 (20200801); A63B2053/0491 (20130101); A63B60/02 (20151001)

Field of Classification Search

CPC: A63B (53/04); A63B (53/0466); A63B (53/0412); A63B (53/0433); A63B (2053/0491); A63B (60/00); A63B (60/02)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5516107	12/1995	Okumoto et al.	N/A	N/A
6056649	12/1999	Imai	N/A	N/A
6254494	12/2000	Hasebe et al.	N/A	N/A
6524198	12/2002	Takeda	N/A	N/A
6607452	12/2002	Helmstetter et al.	N/A	N/A
6776723	12/2003	Bliss et al.	N/A	N/A
6939247	12/2004	Schweigert et al.	N/A	N/A
6991558	12/2005	Beach	473/324	A63B 60/00
7137905	12/2005	Kohno	N/A	N/A
7163468	12/2006	Gibb et al.	N/A	N/A
7294065	12/2006	Liang et al.	N/A	N/A
7413520	12/2007	Hocknell et al.	N/A	N/A
7438649	12/2007	Ezaki et al.	N/A	N/A
7448964	12/2007	Schweigert et al.	N/A	N/A
7500926	12/2008	Rae et al.	N/A	N/A
7575524	12/2008	Willet et al.	N/A	N/A
7651410	12/2009	Shimazaki	N/A	N/A
7731603	12/2009	Beach	N/A	N/A
7762906	12/2009	Murphy et al.	N/A	N/A
7850544	12/2009	Meyer et al.	N/A	N/A
8025591	12/2010	De La Cruz et al.	N/A	N/A
8083609	12/2010	Burnett et al.	N/A	N/A
8100781	12/2011	Burnett et al.	N/A	N/A
8192304	12/2011	Rae et al.	N/A	N/A
8206244	12/2011	Honea et al.	N/A	N/A
8241143	12/2011	Albertsen et al.	N/A	N/A
8267808	12/2011	De La Cruz et al.	N/A	N/A
8333668	12/2011	De La Cruz et al.	N/A	N/A
8371957	12/2012	Schweigert et al.	N/A	N/A

8435137	12/2012	Hirano	N/A	N/A
8496544	12/2012	Curtis et al.	N/A	N/A
8506419	12/2012	Hirano	N/A	N/A
8647216	12/2013	Beach et al.	N/A	N/A
8663029	12/2013	Beach et al.	N/A	N/A
8758153	12/2013	Sargent et al.	N/A	N/A
8790196	12/2013	Solheim et al.	N/A	N/A
8808108	12/2013	Schweigert	N/A	N/A
8821312	12/2013	Burnett et al.	N/A	N/A
8900069	12/2013	Beach et al.	N/A	N/A
9144722	12/2014	Schweigert et al.	N/A	N/A
9168429	12/2014	Schweigert	N/A	N/A
9186561	12/2014	Schweigert	N/A	A63B 53/02
9199138	12/2014	Willett et al.	N/A	N/A
9358430	12/2015	Greaney et al.	N/A	N/A
9643065	12/2016	Stites et al.	N/A	N/A
9675851	12/2016	Schweigert et al.	N/A	N/A
9764205	12/2016	Schweigert et al.	N/A	N/A
9764206	12/2016	Schweigert et al.	N/A	N/A
10080933	12/2017	Stokke	N/A	A63B 53/02
11318359	12/2021	Schweigert et al.	N/A	N/A
2003/0032500	12/2002	Nakahara et al.	N/A	N/A
2005/0239576	12/2004	Stites	473/345	A63B 60/00
2007/0049398	12/2006	Oyama	473/319	A63B 60/00
2009/0017938	12/2008	Yokota	473/349	A63B 53/0466
2009/0029795	12/2008	Schweigert et al.	N/A	N/A
2009/0137338	12/2008	Kajita	473/345	A63B 71/146
2010/0022321	12/2009	Jertson	N/A	N/A
2010/0216570	12/2009	Beach	473/345	A63B 60/02
2010/0234125	12/2009	Aoyama et al.	N/A	N/A
2010/0234126	12/2009	Cackett et al.	N/A	N/A
2011/0250986	12/2010	Schweigert	473/409	A63B 60/02
2012/0071267	12/2011	Burnett et al.	N/A	N/A
2012/0071268	12/2011	Albertsen et al.	N/A	N/A
2012/0083361	12/2011	Beach et al.	N/A	N/A
2013/0005501	12/2012	Ferguson	473/335	A63B 60/02
2013/0102414	12/2012	De la Cruz	473/324	A63B 60/00
2014/0274460	12/2013	Schweigert	473/345	A63B 53/0466
2015/0367196	12/2014	Schweigert	473/342	A63B 53/04
2017/0312595	12/2016	Stokke	N/A	A63B 53/08

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2340875	12/2010	EP	N/A
S57-157373	12/1981	JP	N/A
H08-24376	12/1995	JP	N/A
H11-347157	12/1998	JP	N/A
2005-130935	12/2004	JP	N/A
2005-278800	12/2004	JP	N/A
2009061264	12/2007	JP	N/A

OTHER PUBLICATIONS

US 8,277,335 B2, 10/2012, Beach et al. (withdrawn) cited by applicant

International Search Report / Written Opinion from PCT Application No. PCT/US2014/028134, mailing date Jul. 1, 2014. cited by applicant

International Search Report and Written Opinion for PCT/US2017/043142, mailed on Sep. 27, 2017. cited by applicant

Primary Examiner: Simms, Jr.; John E

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of U.S. patent application Ser. No. 17/661,903, filed May 3, 2022, which is a continuation of U.S. patent application Ser. No. 17/094,707, filed on Nov. 10, 2020, now U.S. Pat. No. 11,318,359, issued on May 3, 2022, which is a continuation of U.S. patent application Ser. No. 16/503,322, filed Jul. 3, 2019, now U.S. Pat. No. 10,857,435, issued on Dec. 8, 2020, which is a continuation of U.S. patent application Ser. No. 15/680,414, filed on Aug. 18, 2017, now U.S. Pat. No. 10,357,700, issued on Jul. 23, 2019, which is a continuation of U.S. patent application Ser. No. 14/836,732, filed on Aug. 26, 2015, now U.S. Pat. No. 9,764,205, issued Sep. 19, 2017, which is a continuation of U.S. patent application Ser. No. 13/804,917, filed on Mar. 14, 2013, now U.S. Pat. No. 9,168,429, issued on Oct. 27, 2015, the contents of which are incorporated fully herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates generally to sports equipment, and relates, more particularly, to golf club heads with optimized characteristics and related methods.

BACKGROUND

(2) Golf club heads often comprise different features that can be designed or configured to improve one or more of their performance characteristics. Innate interplay between such different features often exists, however, such that adjusting or configuring one feature may inherently alter another feature, often disadvantageously. As an example, expanding the strikeface of a golf club to provide a greater impact area can alter the location of the center of gravity of the golf club disadvantageously, and unintended performance consequences may ensue if features are not configured or designed in a balanced manner to account for the interplay between the different features.

(3) Considering the above, further developments with respect to golf club features that are balanced with respect to each other will enhance the performance of golf clubs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure may be better understood from a reading of the following detailed description of examples of embodiments, taken in conjunction with the accompanying figures.
- (2) FIG. 1 illustrates a front view of a golf club head in accordance with the present disclosure.
- (3) FIG. 2 illustrates a side cross-sectional view of the golf club head along line II-II of FIG. 1.
- (4) FIG. 3 illustrates a bottom view of the golf club head of FIGS. 1-2.
- (5) FIG. 4 illustrates a flowchart for a method that can be used to provide, form, and/or

manufacture a golf club head in accordance with the present disclosure.

(6) FIG. 5 illustrates a side view of the golf club head of FIGS. 1-3.

(7) For simplicity and clarity of illustration, the drawing figures illustrate the general manner of construction, and descriptions and details of well-known features and techniques may be omitted to avoid unnecessarily obscuring the present disclosure. Additionally, elements in the drawing figures are not necessarily drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help improve understanding of embodiments of the present disclosure. The same reference numerals in different figures denote the same elements.

(8) The terms “first,” “second,” “third,” “fourth,” and the like in the description and in the claims, if any, are used for distinguishing between similar elements and not necessarily for describing a particular sequential or chronological order. It is to be understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments described herein are, for example, capable of operation in sequences other than those illustrated or otherwise described herein. Furthermore, the terms “include,” and “have,” and any variations thereof, are intended to cover a non-exclusive inclusion, such that a process, method, system, article, device, or apparatus that comprises a list of elements is not necessarily limited to those elements, but may include other elements not expressly listed or inherent to such process, method, system, article, device, or apparatus.

(9) The terms “left,” “right,” “front,” “back,” “top,” “bottom,” “over,” “under,” and the like in the description and in the claims, if any, are used for descriptive purposes and not necessarily for describing permanent relative positions. It is to be understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments of the apparatus, methods, and/or articles of manufacture described herein are, for example, capable of operation in other orientations than those illustrated or otherwise described herein.

(10) The terms “couple,” “coupled,” “couples,” “coupling,” and the like should be broadly understood and refer to connecting two or more elements, mechanically or otherwise. Coupling (whether mechanical or otherwise) may be for any length of time, e.g., permanent or semi-permanent or only for an instant.

(11) The absence of the word “removably,” “removable,” and the like near the word “coupled,” and the like does not mean that the coupling, etc. in question is or is not removable.

(12) As defined herein, two or more elements are “integral” if they are comprised of the same piece of material. As defined herein, two or more elements are “non-integral” if each is comprised of a different piece of material.

DESCRIPTION

(13) In one example, a golf club head can comprise a head body comprising a head interior and a hosel structure. The head interior can be bounded by a head front portion, a head rear portion, a head heel portion, a head toe portion, a head top portion, and a head sole portion. The hosel structure can have a bore for receiving a golf club shaft, where the bore can have a hosel axis. The golf club head can also comprise a head center of gravity, a head horizontal axis extending through the head center of gravity, from the head heel portion to the head toe portion, and parallel to a ground plane when the golf club head is at an address position over the ground plane, a hosel moment of inertia about the hosel axis, and a horizontal moment of inertia about the head horizontal axis. The horizontal moment of inertia can be greater than or equal to 39% of the hosel moment of inertia.

(14) In one example, a golf club head can comprise a head body comprising a head interior and a hosel structure. The head interior can be bounded by a head front portion, a head rear portion, a head heel portion, a head toe portion, a head top portion, and a head sole portion. The hosel structure can have a bore for receiving a golf club shaft, where the bore can have a hosel axis. The golf club head can also comprise a head center of gravity, a head vertical axis extending through

the head center of gravity, from the head top portion to the head sole portion, and perpendicular to a ground plane when the golf club head is at address over the ground plane, a hosel moment of inertia about the hosel axis, and a vertical moment of inertia about the head vertical axis. The vertical moment of inertia can be greater than or equal to 59% of the hosel moment of inertia.

(15) In one implementation, a method for providing a golf club head can comprise providing a head body having a head interior and a hosel structure. The head interior can be bounded by a head front portion, a head rear portion, a head heel portion, a head toe portion, a head top portion, and a head sole portion. The hosel structure can have a bore for receiving a golf club shaft, where the bore can have a hosel axis. The method can also comprise coupling the golf club shaft to the hosel structure. A head horizontal axis can extend through a head center of gravity of the golf club head, from the head heel portion to the head toe portion, and parallel to a ground plane when the golf club head is at an address position over the ground plane. A head vertical axis can extend through the head center of gravity, from the head top portion to the head sole portion, and perpendicular to the ground plane when the golf club head is at address over the ground plane. In addition, providing the head body can comprise at least one of: (a) establishing a horizontal moment of inertia about the head horizontal axis to be greater than or equal to 39% of a hosel moment of inertia about the hosel axis, or (b) establishing a vertical moment of inertia about the head vertical axis to be greater than or equal to 59% of the hosel moment of inertia about the hosel axis.

(16) In one example, a golf club head can comprise a head body comprising a head front portion, a head rear portion, a head heel portion, a head toe portion, a head sole portion, a head top portion, and a hosel structure having a bore for receiving a golf club shaft, where the bore can have a hosel axis. The golf club head can also comprise a strikeface at the head front portion and comprising a strikeface centerpoint, a head volume measured in cc's and comprising a head volume magnitude greater than 420, a head center of gravity, and an optimization characteristic. When the golf club head is at an address position over a ground plane, a head vertical axis extends through the head center of gravity and is orthogonal to the ground plane, and a head horizontal axis extends through the head center of gravity, and is orthogonal to the head vertical axis. A loft plane of the golf club head can be tangent to the strikeface centerpoint. A front plane of the golf club head can extend through the strikeface centerpoint and parallel to the hosel axis. A head depth plane can extend through the strikeface centerpoint, parallel to the head horizontal axis and perpendicular to the loft plane. A CG height axis extends through the head center of gravity and can intersect the head depth plane perpendicularly at a first intersection point. A head CG height of the head center of gravity can be measured, along the CG height axis, between the head center of gravity and the first intersection point. A head CG depth of the head center of gravity can be measured, parallel to the ground plane and orthogonal to the front plane, between (a) a second intersection point located at an intersection between the front plane and the ground plane, and (b) a third intersection point located at an intersection between the head vertical axis and the ground plane. The optimization characteristic can be defined by (a) the head volume magnitude added to (b) a ratio between the head CG depth divided by an absolute value of the head CG height. The optimization characteristic can be greater than or equal to 425.

(17) In one example, a golf club head can comprise a head body comprising a head front portion, a head rear portion, a head heel portion, a head toe portion, a head sole portion, a head top portion, and a hosel structure having a bore for receiving a golf club shaft, where the bore can have a hosel axis. The golf club head can also comprise a strikeface at the head front portion and comprising a strikeface centerpoint, and a head center of gravity. When the golf club head is at an address position over a ground plane, a head vertical axis extends through the head center of gravity and is orthogonal to the ground plane, and a head horizontal axis extends through the head center of gravity, and is orthogonal to the head vertical axis. A loft plane of the golf club head can be tangent to the strikeface centerpoint. A front plane of the golf club head can extend through the strikeface centerpoint and parallel to the hosel axis. A head depth plane can extend through the strikeface

centerpoint, parallel to the head horizontal axis and perpendicular to the loft plane. A CG height axis can extend through the head center of gravity and can intersect the head depth plane perpendicularly at a first intersection point. A head CG height of the head center of gravity can be measured, along the CG height axis, between the head center of gravity and the first intersection point. A head CG depth of the head center of gravity can be measured, parallel to the ground plane and orthogonal to the front plane, between (a) a second intersection point located at an intersection between the front plane and the ground plane, and (b) a third intersection point located at an intersection between the head vertical axis and the ground plane. An absolute value of the head CG height can be less than or equal to 2.54 mm. The head CG depth can be greater than or equal to 40.64 mm.

(18) In one implementation, a method for providing a golf club head can comprise providing a head body comprising a head front portion, a head rear portion, a head heel portion, a head toe portion, a head sole portion, a head top portion, and a hosel structure having a bore for receiving a golf club shaft, the bore having a hosel axis. The method can also comprise coupling a strikeface at the head front portion, and establishing an optimization characteristic of the golf club head. The strikeface comprises a strikeface centerpoint. A head volume of the golf club head can be measured in cc's and can comprise a head volume magnitude greater than 420. When the golf club head is at an address position over a ground plane, a head vertical axis can extend through the head center of gravity and can be orthogonal to the ground plane. A head horizontal axis can extend through the head center of gravity, and can be orthogonal to the head vertical axis. A loft plane of the golf club head can be tangent to the strikeface centerpoint. A front plane of the golf club head can extend through the strikeface centerpoint and parallel to the hosel axis. A head depth plane can extend through the strikeface centerpoint, parallel to the head horizontal axis and perpendicular to the loft plane. A CG height axis can extend through the head center of gravity and can intersect the head depth plane perpendicularly at a first intersection point. A head CG height of the head center of gravity can be measured, along the CG height axis, between the head center of gravity and the first intersection point. A head CG depth of the head center of gravity can be measured, parallel to the ground plane and orthogonal to the front plane, between (a) a second intersection point located at an intersection between the front plane and the ground plane, and (b) a third intersection point located at an intersection between the head vertical axis and the ground plane. The optimization characteristic can be established by (a) the head volume magnitude added to (b) a ratio between the head CG depth divided by an absolute value of the head CG height, where the optimization characteristic can be greater than or equal to 425.

(19) In one example, a golf club head can comprise a head body, a face portion, and a head center of gravity, and at least one of a first performance characteristic or a second performance characteristic. The head body can comprise a head front portion, a head rear portion, a head heel portion, a head toe portion, a head sole portion, a head top portion, and a hosel structure having a bore for receiving a golf club shaft, where the bore can have a hosel axis. The face portion can be at the head front portion and can comprise a strikeface centerpoint, a strikeface perimeter, and a face height bounded by the strikeface perimeter. When the golf club head is at an address position over a ground plane, a head vertical axis extends through the head center of gravity and is orthogonal to the ground plane, and a head horizontal axis extends through the head center of gravity, and is orthogonal to the head vertical axis. A loft plane of the golf club head can be tangent to the strikeface centerpoint. A front plane of the golf club head can extend through the strikeface centerpoint and parallel to the hosel axis. A head depth plane can extend through the strikeface centerpoint, parallel to the head horizontal axis and perpendicular to the loft plane. A CG height axis can extend through the head center of gravity and can intersect the head depth plane perpendicularly at a first intersection point. A head CG height of the head center of gravity can be measured, along the CG height axis, between the head center of gravity and the first intersection point. A head CG depth of the head center of gravity can be measured, parallel to the ground plane

and orthogonal to the front plane, between (a) a second intersection point located at an intersection between the front plane and the ground plane, and (b) a third intersection point located at an intersection between the head vertical axis and the ground plane. The face height can be approximately 33 mm to approximately 71 mm, measured parallel to the loft plane. The first performance characteristic can comprise the head CG height being less than or equal to approximately 5.08 mm. The second performance characteristic can comprise a CG performance ratio of less than or equal to 0.56, as defined by (a) 76.2 mm minus the face height, divided by (b) the head CG depth.

(20) In one example, a golf club head can comprise a head body, a face portion, and a head center of gravity. The head body can comprise a head front portion, a head rear portion, a head heel portion, a head toe portion, a head sole portion, a head top portion, and a hosel structure having a bore for receiving a golf club shaft, the bore having a hosel axis. The face portion can be coupled to the head front portion and can comprise a strikeface having a strikeface centerpoint, a strikeface perimeter, and a face height. When the golf club head is at an address position over a ground plane, a head vertical axis extends through the head center of gravity and is orthogonal to the ground plane, and a head horizontal axis extends through the head center of gravity, and is orthogonal to the head vertical axis. A loft plane of the golf club head can be tangent to the strikeface centerpoint. A front plane of the golf club head can extend through the strikeface centerpoint and parallel to the hosel axis. A head depth plane can extend through the strikeface centerpoint, parallel to the head horizontal axis and perpendicular to the loft plane. A CG height axis can extend through the head center of gravity and can intersect the head depth plane perpendicularly at a first intersection point. A head CG height of the head center of gravity can be measured, along the CG height axis, between the head center of gravity and the first intersection point. A head CG depth of the head center of gravity can be measured, parallel to the ground plane and orthogonal to the front plane, between (a) a second intersection point located at an intersection between the front plane and the ground plane, and (b) a third intersection point located at an intersection between the head vertical axis and the ground plane. The face height can be approximately 33 mm to approximately 71 mm, as delimited by the strikeface perimeter and measured parallel to the loft plane. A CG performance ratio between (a) 76.2 mm minus the face height, and (b) the head CG depth, is less than or equal to 0.56. The head body can comprise a driver-type body. A head volume of the golf club head can be approximately 420 cc to approximately 470 cc. A head weight of the golf club head can be approximately 185 grams to approximately 225 grams. The head CG height can be approximately 0 mm to approximately 3.18 mm. The head CG depth can be approximately 25 mm to approximately 102 mm. The head body can comprise a weight structure located towards the sole portion and the rear portion of the head body.

(21) In one implementation, a method for providing a golf club head can comprise providing a head body having a head front portion, a head rear portion, a head heel portion, a head toe portion, a head sole portion, a head top portion, and a hosel structure having a bore for receiving a golf club shaft, the bore having a hosel axis. The method can also comprise coupling a face portion to the head front portion, the face portion comprising a strikeface having a strikeface centerpoint, a strikeface perimeter, and a face height bounded by the strikeface perimeter. The method can further comprise establishing at least one of: a first performance characteristic of the golf club head, or a second performance characteristic of the golf club head. When the golf club head is at an address position over a ground plane, a head vertical axis extends through a head center of gravity of the golf club head and is orthogonal to the ground plane, and a head horizontal axis extends through the head center of gravity, and is orthogonal to the head vertical axis. A loft plane of the golf club head can be tangent to the strikeface centerpoint. A front plane of the golf club head can extend through the strikeface centerpoint and parallel to the hosel axis. A head depth plane can extend through the strikeface centerpoint, parallel to the head horizontal axis and perpendicular to the loft plane. A CG height axis can extend through the head center of gravity and can intersect the head

depth plane perpendicularly at a first intersection point. A head CG height of the golf club head can be measured, along the CG height axis, between the head center of gravity and the first intersection point. A head CG depth of the head center of gravity can be measured, parallel to the ground plane and orthogonal to the front plane, between: (a) a second intersection point located at an intersection between the front plane and the ground plane, and (b) a third intersection point located at an intersection between the head vertical axis and the ground plane. The face height can be approximately 33 mm to approximately 71 mm, measured parallel to the loft plane. The first performance characteristic can comprise the head CG height being less than or equal to approximately 5.08 mm. The second performance characteristic can comprise a CG performance ratio of less than or equal to 0.56, as defined by (a) 76.2 mm minus the face height, divided by (b) the head CG depth.

(22) Other examples and embodiments are further disclosed herein. Such examples and embodiments may be found in the figures, in the claims, and/or in the present description.

(23) Turning to the drawings, FIG. 1 illustrates a front view of golf club head **1000**, comprising head body **1100** and face portion **1200**. Face portion **1200** includes strikeface **1210**. FIG. 2 illustrates a side cross-sectional view of golf club head **1000** along line II-II of FIG. 1. FIG. 3 illustrates a bottom view of golf club head **1000**. FIGS. 1-3 present golf club head **1000** at an address position relative to ground plane **1010**, where hosel axis **1710** is at a 60-degree angle with ground plane **1010** with respect to a front view of golf club head **1000** (FIG. 1), and where hosel axis **1710** is substantially orthogonal to ground plane **1010** with respect to a side view of golf club head **1000** (FIG. 2).

(24) In the present embodiment, head body **1100** and face portion **1200** comprise separate pieces of material coupled together, for example, via a welding process. In other examples, however, face portion **1200** may comprise a single piece of material with one or more portions of head body **1100**, such as head front portion **1110**, head top portion **1120**, head sole portion **1130**, head heel portion **1140**, head toe portion **1150**, and/or head rear portion **2160**. Head forward surface **1160** of golf club head **1000** comprises strikeface **1210**, face portion **1200**, and at least part of head front portion **1110**. In some embodiments, head forward surface **1160** also can include at least part of head sole portion **1130**. In the same or different embodiments, head front portion **1110** can include strikeface **1210** and/or face portion **1200**.

(25) Face portion **1200** comprises strikeface **1210** having strikeface centerpoint **1211**, strikeface perimeter **1212**, and face height **1213**. Strikeface centerpoint **1211** is located at a geometric centerpoint of strikeface perimeter **1212** in the present example, and at a midpoint of face height **1213**. In the same or other examples, strikeface centerpoint **1211** also can be centered with respect to engineered impact zone **1250**, which can be defined by a region of grooves **1259** of strikeface **1210**. As another approach, strikeface centerpoint **1211** can be located in accordance with the definition of a golf governing body such as the United States Golf Association (USGA). For example, strikeface centerpoint **1211** can be determined in accordance with Section 6.1 of the USGA's Procedure for Measuring the Flexibility of a Golf Clubhead (USGA-TPX3004, Rev. 1.0.0, May 1, 2008) (available at <http://www.usga.org/equipment/testing/protocols/Procedure-For-Measuring-The-Flexibility-Of-A-Golf-Club-Head/>) (the "Flexibility Procedure").

(26) Golf club head **1000** comprises loft plane **2270** (FIG. 2), which is at least tangent to strikeface centerpoint **1211** at strikeface **1210**. Face height **1213** can be measured parallel to loft plane **2270** between strikeface top end **1215** and strikeface bottom end **1216** of strikeface perimeter **1212**, and can be of approximately 33 millimeters (mm) to approximately 71 mm in the present or other examples.

(27) Strikeface perimeter **1212**, comprising strikeface top end **1215** and strikeface bottom end **1216** defining face height **1213**, need not bound an entirety of face portion **1200**. For example, as seen in FIG. 1, strikeface **1210** is bounded by strikeface perimeter **1212** and is only part of face portion **1200**. In some examples, strikeface **1210** can comprise a roll radius and/or a bulge radius, and

strikeface perimeter **1212** can be defined along a transition boundary where a contour of face portion **1200** departs from the roll radius and/or the bulge radius of strikeface **1210**. For example, FIG. 2 contains a zoom view of part of the top transition boundary of golf club head **1000**, highlighting vertical roll radius **2170** extending along strikeface **1210**, and showing how strikeface top end **1215** is located at the top transition boundary where head forward surface **1160** departs from vertical roll radius **2170**. FIG. 2 also contains a zoom view of part of the bottom transition boundary of golf club head **1000**, highlighting vertical roll radius **2170** extending vertically along strikeface **1210**, and showing how strikeface bottom end **1216** is located at the bottom transition boundary where head forward surface **1160** departs from vertical roll radius **2170**.

(28) In the same or other embodiments, strikeface perimeter **1212** can be defined with respect to the edge of a strikeplate comprising the strikeface. For instance, face portion **1200** comprises strikeplate **1220**, where strikeface **1210** forms an exterior surface of faceplate **1220**, and where strikeplate **1220** is joined to head front portion **1110** along strikeplate edge **1221**. In the present example, strikeplate edge **1221** defines at least part of strikeface perimeter **1212**, including the top and bottom sections of strikeface perimeter **1212** where strikeface top end **1215** and strikeface bottom end **1216** are respectively located to define face height **1213**, but there can be other examples where the strikeplate edge of the strikeplate can define a majority or all of the strikeface perimeter of the strikeface.

(29) As shown in FIG. 2, golf club head **1000** also comprises head center of gravity (CG) **2500**, head depth plane **2310**, and CG height axis **2320**, where head depth plane **2310** extends through strikeface centerpoint **1211** and is perpendicular to loft plane **2270**, and where CG height axis **2320** extends through head center of gravity **2500** and intersects head depth plane **2310** perpendicularly at intersection point **2801**.

(30) Head center of gravity **2500** comprises CG height **2520** and CG depth **2510**, which locate head center of gravity **2500** relative to golf club head **1000**. In the present example, CG height **2520** can be measured along CG height axis **2320**, between head center of gravity **2500** and intersection point **2801**. CG depth **2510** can be measured, as seen in FIG. 2, parallel to ground plane **1010** and between intersection points **2802-2803**. In the present example, intersection point **2802** is defined by the intersection between ground plane **1010** and front plane **2280**, where front plane **2280** extends through strikeface centerpoint **1211**, is parallel to hosel axis **1710**, and is orthogonal to ground plane **1010** when golf club head **1000** is at the address position. In addition, intersection point **2803** is defined by the intersection between ground plane **1010** and head vertical axis **1610**, where head vertical axis **1610** extends through head center of gravity **2500**, and is orthogonal to ground plane **1010** when golf club head **1000** is at the address position. Head center of gravity **2500** can also be located relative to ground plane **1010**, where head CG elevation **2530** of head center of gravity **2500** can be measured along head vertical axis **1610**, between weight center **2750** and ground plane **1010**.

(31) Head body **1100** of golf club head **1000** also comprises hosel structure **1217** (FIG. 1) and hosel axis **1710** extending along a center of a bore of hosel structure **1217**. In the present example, a hosel coupling mechanism of golf club head **1000** comprises hosel structure **1217** and shaft sleeve **1411**, where shaft sleeve **1411** can be coupled to an end of golf shaft **1410**. Shaft sleeve **1411** can couple with hosel structure **1217** in a plurality of configurations, thereby permitting golf shaft **1410** to be secured to hosel structure **1217** at a plurality of angles relative to hosel axis **1710**. There can be other examples, however, where shaft **1410** can be non-adjustably secured to hosel structure **1217**.

(32) Golf club heads in accordance with the present disclosure can be configured to exhibit one or more optimization characteristics that optimize or balance the performance thereof. For example, one characteristic of golf club heads that the present designs strive to optimize is that of face height and/or face size. Maximizing the face height and/or face size of a golf club head can have several benefits, such as increasing the target impact area of the strikeface to yield a more forgiving club

head that produces better results for golf shots that are hit off-center of the strikeface centerpoint. In addition, a strikeface of larger height and/or size can provide for better energy transfer to the golf ball upon impact therewith, and may thus increase a characteristic time or a “spring effect” of the golf club head to achieve golf shots of longer distance. In some examples, the height or size of the strikeface may be augmented to achieve a characteristic time limit set by a golf governing body, such as the characteristic time limit of 239 microseconds (μs) set by the USGA in its Flexibility Procedure.

(33) Indiscriminately increasing face height and/or size, however, can adversely affect performance in other areas, such as with respect to launch angle, ball spin, and/or ball speed of the golf ball upon impact with the strikeface. For example, increasing face height and/or size can decrease the CG depth between the center of gravity and the strikeface centerpoint of the golf club head to bring the center of gravity forward, thereby reducing the dynamic loft of the golf club head and thus decreasing the launch angle for the golf ball. As another example, increasing face height and/or size can raise the CG height between the center of gravity and the head depth plane to elevate the center of gravity away from the sole of the golf club head, thereby inhibiting a gear effect between the strikeface and the golf ball, thus preventing the golf club head from decreasing the amount of backspin of the golf ball created upon impact, and thus decreasing the distance the ball will travel due to the backspin.

(34) Considering the above, the height or size of the face of the golf club head should be balanced with respect to the location of the center of gravity. With respect to golf club head **1000**, strikeface **1210** has been increased to comprise an augmented face size and/or face height **1213** to provide a larger impact area and greater energy transfer to golf ball **2900**. In particular, golf club head **1000** can be configured so that face height **1213** can be of approximately 33 mm to approximately 71 mm to provide greater impact area and energy transfer upon impact with golf ball **2900**. In some examples, an area of strikeface **1210**, including the augmented face area, can be of approximately 23.6 centimeters squared (cm^2) to approximately 45.2 cm^2 .

(35) Notwithstanding the augmented face size and/or face height **1213** described above, golf club head **1000** still restricts CG height **2520** from increasing towards head top portion **1120** and/or from straying too far from head depth plane **2310**. For example, golf club head **1000** comprises a first optimization characteristic satisfying Relation 1 below:

$$|\text{CG height.sub.2520}| \leq 5.08 \text{ mm} \quad [\text{Relation 1}]$$

There can be examples where CG height **2520** can be of approximately 0 mm up to Relation 1's limit of 5.08 mm. CG height **2520** can also be of up to a maximum of approximately 4.45 mm, 3.81 mm, or 3.18 mm in other examples. In some implementations, the first optimization characteristic can decrease the backspin of golf ball **2900** via a gear effect between strikeface **1210** and golf ball **2900** for better performance. Although head center of gravity **2500** is shown in FIG. 2 as being below depth plane **2310**, such that CG height **2520** extends between depth plane **2310** and head sole portion **1130**, there can be embodiments where head center of gravity **2500** can be above depth plane **2310**, such that CG height **2520** extends between depth plane **2310** and head top portion **1120**, while still satisfying Relation 1 above.

(36) In addition, and considering the augmented face size and/or face height **1213** described above, golf club head **1000** still restricts center of gravity **2500** from moving towards strikeface **1212**, thereby preventing CG depth **2510** from unduly decreasing. For example, golf club head **1000** comprises a second optimization characteristic satisfying Relation 2 below:

$$(37) \quad \frac{76.2\text{mm} - \text{faceheight}_{1213}}{\text{CGdepth}_{2510}} \leq 0.56 \quad [\text{Relation 2}]$$

Accordingly, the relationship between face height **1213** and CG depth **2510** is balanced pursuant to Relation 2 to maintain the second optimization characteristic of less than or equal to 0.56, thereby limiting the amount that CG depth **2510** can decrease towards strikeface **1210**. There can be examples where CG depth **2510** can be approximately 25 mm to approximately 102 mm. In the

same or other examples, CG depth **2510** can be at least approximately 39 mm. In some implementations, the second optimization characteristic can increase or optimize at least one of a dynamic loft of golf club head **1000** or a launch angle of golf ball **2900** upon impact therebetween. (38) In some examples, golf club head **1000** may be configured to comprise only one of the first or second optimization characteristics described above. For example, golf club head **1000** may comprise the first optimization characteristic and not the second optimization characteristic, thus satisfying Relation 1 without having to satisfy Relation 2. As another example, golf club head **1000** may comprise the second optimization characteristic and not the first optimization characteristic, thus satisfying Relation 2 without having to satisfy Relation 1. In addition, there can be embodiments where golf club head **1000** satisfies both Relations 1 and 2, and thus comprises the first and second optimization characteristics.

(39) Golf club head **1000** can also comprise a third optimization characteristic with respect to head volume (HV) **2600** thereof. In the present example, head body **1000** of golf club head **1000** comprises a driver-type body with a head volume greater than or equal to 420 cubic centimeters (cc), and thus has a head volume magnitude greater than or equal to 420. For example, head body **1000** can comprise a head volume of 420 cc, thus having a head volume magnitude of 420. As another example, golf club head **1000** can comprise a head volume of 460 cc, thus having a head volume magnitude of 460. Golf club head **1000** can comprise a head volume up to approximately 470 cc, in some implementations, and/or a total head weight of approximately 185 grams to approximately 225 grams. In some specific examples, the total head weight can be approximately 202 grams, and/or the head volume can be of approximately 460 cc.

(40) The third optimization characteristic can control the relationship between head volume **2600** and the location of center of gravity **2500**, and can be defined to satisfy Relation 3 below:

$$(41) \quad HV + \frac{CG_{depth_{2510}}}{.Math. \cdot CG_{height_{2520}} \cdot .Math.} \geq 425 \quad [\text{Relation3}]$$

In some instances, head volume **2600** can be increased to adjust, for instance, a moment of inertia (MOI) of golf club head **1000**. An unrestrained increase in head volume, however, can have detrimental effects with respect to other characteristics of the golf club head. For example, increasing head volume **2600** can cause head center of gravity **2500** to shift towards head front portion **1110**, towards head top portion **1120**, towards other undesired directions, and/or away from a desired center of gravity location or direction(s), thereby hampering the performance of golf club head **1000**. Such undesired changes in the center of gravity location can detrimentally affect one or more characteristics of the golf club head, such as launch speed, launch angle, gear effect, backspin, and or shot distance. Accordingly, the third optimization characteristic used to balance the relationship between head volume **2600** and the location of head center of gravity **2500** can be established to yield desirable and balanced attributes for golf club head **1000**. For instance, a weight distribution of golf club head **1000** can be configured to satisfy Relation 3 so that golf club head **1000** can exhibit the third optimization characteristic, thereby permitting head volume **2600** to be augmented for greater moment of inertia and greater energy transfer to golf ball **2900** upon a golf impact between strikeface **1210** and golf ball **2900**. In the same or other implementations, the weight distribution of golf club head **1000** can be configured for restricting CG depth **2510** from decreasing towards head front portion **1110** due to the augmented head volume **2600**, thus increasing at least one of a dynamic loft of strikeface **1210** or a launch angle of golf ball **2900** upon the golf impact. In addition, the weight distribution of golf club head **1000** can be configured for restricting CG height **2520** from increasing towards head top portion **1120** as a result of the augmented head volume **2600**, thereby decreasing a backspin of golf ball **2900** via a gear effect between strikeface **1210** and golf ball **2900** upon the golf impact.

(42) Considering the above, to attain the third optimization characteristic in compliance with Relation 3, CG depth **2510** can be configured to be greater than or equal to 40.64 mm. In the same or other embodiments, the absolute value of CG height **2520** can be less than or equal to 2.54 mm.

Note that CG height **2500** is characterized as an absolute value, considering that head center of gravity **2500** can be above or below head depth plane **2310** in some embodiments. Although the third optimization characteristic has a lower bound of at least 425, there can be other embodiments where the third optimization characteristic can be defined with respect to other lower bounds. For instance, the third optimization characteristic can comprise a lower bound of at least 435 or 445 in some implementations. The location of head center of gravity **2500** can also be designed or configured with respect to other features of golf club head **1000** in order to satisfy Relation 3 and/or to attain the third optimization characteristic. For instance, the location of head center of gravity **2500** can be configured such that CG depth **2510** comprises between approximately 25% to approximately 80% of head depth length **2312**, where head depth length **2312** is measured from strikeface centerpoint **1211** to an intersection of an exterior of head rear portion **2160** by head depth plane **2310**. As another example, the location of head center of gravity **2500** can be configured such that CG height **2520** comprises between approximately 0% to approximately 13% of CG height axis length **2322**, where CG height axis length **2322** is measured from an intersection of an exterior of head top portion **1120** by CG height axis **2320**, to an intersection of an exterior of head sole portion **1130** by CG height axis **2320**.

(43) Golf club head **1000** also can comprise a fourth optimization characteristic with respect to a balance between hosel MOI **1711** (FIG. 1) and horizontal MOI **1811** (FIG. 1). Hosel MOI **1711** is defined about hosel axis **1710**. Horizontal MOI **1811** is defined about head horizontal axis **1810**, which extends through head center of gravity **2500**, from head heel portion **1140** to head toe portion **1150**, and parallel to ground plane **1010** when golf club head **1000** is at the address position over ground plane **1010**.

(44) In some examples, horizontal MOI **1811** can be increased to restrict a rotation of golf club head **1000** about head horizontal axis **1810** when strikeface **1210** hits golf ball **2600** off-center towards head top portion **1120** or head sole portion **1130**, thereby increasing the forgiveness of golf club head **1000** for such high or low mis-hits. For instance, to increase horizontal MOI **1811**, weight may be added or repositioned towards head front portion **1110** and/or head rear portion **2160**. In the same or other examples, golf club head **1000** can be lengthened towards head front portion **1110** and/or head rear portion **2160**.

(45) Such adjustments or changes to increase horizontal MOI **1811** can be made up to a point, however, before they start affecting other golf club head characteristics. For example, unrestrained adjustments to increase horizontal MOI **1811** can lead to an undue increase in hosel MOI **1711** if not properly balanced, thereby increasing the resistance of golf club head **1000** to rotate about hosel axis **1710**, and thus making it hard for a person to “turn over” the golf club during a golf swing for proper positioning or “squaring” of golf club head **1000** at impact with golf ball **2600**. An increase in hosel MOI **1711** also can restrict or reduce a gearing effect between golf ball **2600** and strikeface **1210** that would otherwise impart some corrective spin to golf ball **2600** during off-center impacts.

(46) To reduce hosel MOI **1711**, golf club head **1000** can be designed to limit the distance between hosel axis **1710** and any additional or discretionary mass of golf club head **1000**. Such approaches to decrease hosel MOI **1711**, if not properly balanced, can be incompatible with some of the approaches described above to increase horizontal MOI **1811**. Accordingly, weight addition or redistribution for golf club head **1000** to increase horizontal MOI **1811** should be balanced with respect to maintaining or restricting an increase in hosel MOI **1711**.

(47) In light of the above, the fourth optimization characteristic of golf club head **1000** controls the relationship between horizontal MOI **1811** and hosel MOI **1711** to satisfy to satisfy Relation 4 below:

(Horizontal MOI.sub.1811) ≥ 39% (Hosel MOI.sub.1711) [Relation 4]

(48) There can be examples where golf club head **1000** can be configured so that its fourth optimization characteristic can surpass the requirements of Relation 4. As an example, in some implementations, the fourth optimization characteristic of golf club head **1000** can be configured so

that horizontal MOI **1811** is greater than or equal to 40% of hosel MOI **1711**, greater than or equal to 45% of hosel MOI **1711**, or greater than or equal to 50% of hosel MOI **1711**. In the present example, horizontal MOI **1811** is approximately 3740 grams-square-centimeter (g.Math.cm.sup.2), but there can be examples where it can range between approximately 2800 g.Math.cm.sup.2 to approximately 4300 g.Math.cm.sup.2. Hosel MOI **1711** is approximately 9370 g.Math.cm.sup.2 in the present example, but can range between approximately 7000 g.Math.cm.sup.2 and approximately 11,000 g.Math.cm.sup.2 in the same or other examples.

(49) Golf club head **1000** also can comprise a fifth optimization characteristic with respect to a balance between hosel MOI **1711** and vertical MOI **1611** (FIG. 1). Vertical MOI **1611** is defined about head vertical axis **1610**, which extends through head center of gravity **2500**, from head top portion **1120** to head sole portion **1130**, and orthogonal to ground plane **1010** when golf club head **1000** is at the address position. Vertical MOI **1611** is approximately 5300 g.Math.cm.sup.2 in the present example, but can range between approximately 4700 g.Math.cm.sup.2 and approximately 6000 g.Math.cm.sup.2 in the same or other examples.

(50) In some examples, vertical MOI **1611** can be increased to restrict a rotation of golf club head **1000** about head vertical axis **1610** when strikeface **1210** hits golf ball **2600** off-center towards head heel portion **1140** or towards head toe portion **1150**, thereby increasing the forgiveness of golf club head **1000** for such heel-side or toe-side mis-hits. For instance, to increase vertical MOI **1611**, weight can be added or repositioned towards head heel portion **1140** and/or head toe portion **1150**. In the same or other examples, the golf club head can be lengthened towards head heel portion **1140** and/or head toe portion **1150**.

(51) Such adjustments or changes to increase vertical MOI **1611** can be made up to a point, however, before they start affecting other golf club head characteristics. For example, unrestrained adjustments to increase vertical MOI **1611** can lead to undue increase in hosel MOI **1711** if not properly balanced, thereby increasing the resistance of golf club head **1000** to rotate about hosel axis **1710** as described above. In addition, some approaches to decrease hosel MOI **1711**, if not properly balanced, can be incompatible with some of the approaches described above to increase vertical MOI **1611**. Accordingly, weight addition or redistribution for golf club head **1000** to increase vertical MOI **1611** should be balanced with respect to maintaining or restricting an increase in hosel MOI **1711**.

(52) In light of the above, the fifth optimization characteristic of golf club head **1000** controls the relationship between vertical MOI **1611** and hosel MOI **1711** to satisfy Relation 5 below:

(Vertical MOI.sub.1611) ≥ 59% (Hosel MOI.sub.1711) [Relation 5]

(53) There can be examples where golf club head **1000** can be configured so that its fifth optimization characteristic can surpass the requirements of Relation 5. As an example, in some implementations, the fifth optimization characteristic of golf club head **1000** can be configured so that vertical MOI **1611** is greater than or equal to 60% of hosel MOI **1711**, greater than or equal to 65% of hosel MOI **1711**, or greater than or equal to 70% of hosel MOI **1711**. In some examples, golf club head **1000** can be configured so that its fourth optimization characteristic satisfies Relation 4, while its fifth optimization characteristic also satisfies Relation 5.

(54) In some implementations, golf club head **1000** can be configured to exhibit the first, second, third, fourth, and/or fifth optimization characteristics described above by adjusting a distribution of mass or a relationship between different elements of golf club head **1000**. To such ends, golf club head **1000** can comprise weight structure **2700**, located towards head sole portion **1130** and head rear portion **2160**, as seen in FIGS. 2-3. In some configurations, weight structure **2700** can be designed and/or located to satisfy the constraints imposed by Relation(s) 1, 2, 3, 4, and/or 5, thereby balancing the face height or size of strikeface **1210**, head volume **2600**, the location of center of gravity **2500**, and/or the different moments of inertia of golf club head **1000**.

(55) As can be seen in FIG. 3, weight structure **2700** can be located relative to clock grid **3500**, which can be aligned with respect to strikeface **1210**. For example, clock grid **3500** comprises 12

o'clock ray **3512**, which is aligned with strikeface centerpoint **1211** in the present embodiment. 12 o'clock ray **3512** is orthogonal to front intersection line **3271**, which is defined by the intersection of loft plane **2270** (FIGS. 2-3) and ground plane **1010** (FIGS. 1-2). Clock grid **3500** can be centered along 12 o'clock ray **3512**, at a midpoint between a front end of front portion **1110** and a rear end of rear portion **2160**. In the same or other examples, clock grid centerpoint **3515** can be centered proximate to a geometric centerpoint of golf club head **1000**. Clock grid **3500** also comprises 3 o'clock ray **3503** extending towards head heel portion **1140**, and 9 o'clock ray **3509** extending towards head toe portion **1150**.

(56) Weight perimeter **2705** of weight structure **2700** is located in the present embodiment towards head rear portion **2160**, at least partially bounded between 4 o'clock ray **3504** and 8 o'clock ray **3508** of clock grid **3500**, while weight center **2750** is located between 5 o'clock ray **3505** and 7 o'clock ray **3507**. In examples such as the present one, weight perimeter **2705** is fully bounded between 4 o'clock ray **3504** and 8 o'clock ray **3508**. Although weight perimeter **2705** is defined external to golf club head **1000** in the present example, there can be other examples where weight perimeter may extend into an interior of, or be defined within, golf club head **1000**. In some examples, the location of weight **2700** can be established with respect to a broader area. For instance, in such examples, weight perimeter **2705** of weight structure **2700** can be located towards head rear portion **2160**, at least partially bounded between 4 o'clock ray **3504** and 9 o'clock ray **3509** of clock grid **3500**, while weight center **2750** can be located between 5 o'clock ray **3505** and 8 o'clock ray **3508**.

(57) In the same or other embodiments, weight structure **2700** can extend or be shifted towards heel portion **1140**. For instance, weight perimeter **2705** and/or weight center **2750** can be shifted towards 4 o'clock ray **3504** than towards 9 o'clock ray **3509**. Biasing weight structure **2700** towards head heel end **1140** can permit a decrease in hosel MOI **1711** about hosel axis **1710** by limiting the distance between hosel axis **1710** and weight structure **2700**, thereby allowing easier turning of golf club head **1000** about hosel axis **1710** during a swing.

(58) In some examples, weight structure **2700** can comprise a mass of approximately 2 grams to approximately 50 grams, and/or a volume of approximately 1 cc to approximately 30 cc. In the present example, weight structure **2700** protrudes from the external contour of head sole portion **1130**, and is thus at least partially external to allow for greater adjustment of head center of gravity **2500**.

(59) Weight structure **2700** can comprise removable weight **2790** in the same or other examples, where removable weight **2790** can comprise a mass of approximately 0.5 grams to approximately 30 grams, and can be replaced with one or more other similar weights to adjust the location of head center of gravity **2500** if needed to satisfy Relation(s) 1, 2, 3, 4, and/or 5. In the same or other examples, weight center **2750** can comprise at least one of a center of gravity of weight structure **2700**, a center of gravity of removable weight **2790**, a geometric center of weight structure **2700**, and/or a geometric center of removable weight **2790**.

(60) Weight center **2750** can be located with respect to ground plane **1010** and weight center elevation axis **2340**, which extends between weight center **2750** and ground plane **1010**. Weight center elevation axis **2340** is orthogonal to ground plane **1010** when golf club head **1000** is at the address position. Weight center elevation **2730** for weight center **2750** can thus be measured along weight center elevation axis **2340**, between weight center **2750** and ground plane **1010**. In addition, weight center depth **2710** for weight center **2750** can be measured, parallel to ground plane **1010**, between intersection points **2802** and **2804**. In the present example, intersection point **2804** is defined by the intersection between ground plane **1010** and weight center elevation axis **2340** when golf club head **1000** is at the address position. Weight center **2750** can be located in the same or other embodiments such that weight distance **2751** (FIG. 2), which separates head center of gravity **2500** from weight center **2750**, can be approximately 25 mm to approximately 102 mm.

(61) There can also be embodiments where face portion **1200** can comprise a reduced thickness,

which may be reinforced as needed with one or more reinforcing structures at the backside of strikeface **1210** and/or at the junction between face portion **1200** and head front portion **1110**. Other mass redistribution mechanisms can be employed as well if desired to satisfy Relation(s) 1, 2, 3, 4, and/or 5.

(62) In some implementations, a relationship or ratio between head center of gravity **2500** and weight center **2750** can be configured to permit one or more of Relation(s) 1, 2, 3, 4, or 5 to be satisfied. For example, an elevation ratio, defined by the ratio of weight center elevation **2730** over head CG elevation **2530**, can be greater than 0.44 to help maintain head center of gravity **2500** closer to head sole portion **1130**. As another example, a depth ratio, defined by the ratio of weight center depth **2710** over head CG depth **2510**, can be less than 2.54 to preventing CG depth **2510** from unduly decreasing towards head front portion **1110**. There can be some implementations where head CG elevation **2530** can be less than approximately 28.5 mm, where weight center elevation **2730** can be less than approximately 12.5 mm, and/or where weight center depth **2710** can be greater than approximately 99.7 mm.

(63) FIG. 4 illustrates a flowchart for method **4000**, which can be used to provide, form, and/or manufacture a golf club head in accordance with the present disclosure. In some examples, the golf club head can be similar to golf club head **1000** (FIGS. 1-3) presented above.

(64) Method **4000** comprises block **4100** for providing a head body of a golf club head comprising a head front portion. In some examples, the head body can be similar to head body **1100** (FIGS. 1-3), and the head front portion can be similar to head front portion **1110** (FIGS. 1-3).

(65) Block **4200** of method **4000** comprises coupling a face portion to the head front portion, the head front portion comprising a strikeface with an augmented face size. In some examples, the face portion can be similar to face portion **1200** (FIGS. 1-2), with strikeface **1210** having the augmented face size described above with respect thereto. For example, the augmented face size of the strikeface may permit its face height to be of up to approximately 71 mm in some examples.

(66) Method **4000** can comprise block **4300** for configuring the golf club head to comprise a first optimization characteristic, where a CG height between a center of gravity of the golf club head and a head depth plane of the golf club head can be approximately 0 mm to approximately 5.08 mm or 0.200 inches. In some examples, the first optimization characteristic can be similar to that described above with respect to Relation 1 for balancing golf club head face height or size with respect to center of gravity height. In some examples, the CG height may be similar to CG height **2520** (FIG. 2); the center of gravity may be similar to head center of gravity **2500** (FIG. 2); and the head depth plane can be similar to head depth plane **2310** (FIG. 2).

(67) There can be implementations where method **4000** can comprise block **4400** for configuring the golf club head to comprise a second optimization characteristic, where a ratio between (a) 76.2 mm (or approximately 3.0 inches) minus the face height and (b) a CG depth between the strikeface centerpoint and the center of gravity, is less than 0.56. In some examples, the second optimization characteristic can be similar to that described above with respect to Relation 2 for balancing golf club head face height or size with respect to center of gravity depth. For example, the face height can be similar to face height **1213**, and the CG depth can be similar to CG depth **2510**.

(68) In some examples, method **4000** can comprise block **4500** for configuring the golf club head to comprise a third optimization characteristic where a head volume magnitude, added to a ratio between the CG depth and the CG height, is greater than or equal to 425. In some implementations, the third optimization characteristic can be similar to that described above with respect to Relation 3 for balancing head volume relative to center of gravity location. For example, the head volume magnitude can be similar to the magnitude of head volume **2600** (FIG. 2), the CG depth can be similar to CG depth **2510**, and the CG height can be similar to CG height **2520**.

(69) Method **4000** can comprise block **4600** in some embodiments for configuring the golf club head to comprise a fourth optimization characteristic, where the golf club head's horizontal moment of inertia is greater than or equal to 39% of its hosel moment of inertia. In some implementations,

the fourth optimization characteristic can be similar to that described above with respect to Relation 4 for balancing horizontal MOI **1811** with respect to hosel MOI **1711** (FIG. 1). In the same or other examples, the magnitude of the horizontal moment of inertia can be similar to that described above with respect to horizontal MOI **1811**. In addition, the magnitude of the hosel moment of inertia can be similar to that described above with respect to hosel MOI **1711**. There can also be examples where the horizontal moment of inertia and/or the hosel moment of inertia can be balanced with respect to other features, such as with respect to a vertical moment of inertia of the golf club head. (70) Block **4700** of method **4000** can be carried out in some implementations for configuring the golf club head to comprise a fifth optimization characteristic, where the golf club head's vertical moment of inertia is greater than or equal to 59% of its hosel moment of inertia. In some implementations, the fifth optimization characteristic can be similar to that described above with respect to Relation 5 for balancing vertical MOI **1611** with respect to hosel MOI **1711** (FIG. 1). In the same or other examples, the magnitude of the vertical moment of inertia can be similar to that described above with respect to vertical MOI **1611**. In addition, the magnitude of the hosel moment of inertia can be similar to that described above with respect to hosel MOI **1711**. There can also be examples where the vertical moment of inertia and/or the hosel moment of inertia can be balanced with respect to other features, such as with respect to the horizontal moment of inertia of block **4500**.

(71) In the present example, method **4000** also comprises block **4800** for providing a mass redistribution mechanism to adjust the center of gravity of the golf club head. In some examples, the mass redistribution mechanism can be configured to permit the golf club head to achieve the requirements of block **4300**, block **4400**, block **4500**, block **4600**, and/or block **4700** of method **4000**. The mass redistribution mechanism can comprise a weight structure, such as weight structure **2700** (FIGS. 2-3), which can adjust the location of the center of gravity towards the sole and/or the rear portion of the golf club head if desired. In the same or other embodiments, the mass redistribution mechanism can comprise a reduced thickness of the face portion of the golf club head, which may be reinforced if needed with one or more reinforcing structures, such as at the backside of the strikeface, and/or at a junction between the face portion and the head body of the golf club head.

(72) In some examples, one or more of the different blocks of method **4000** can be combined into a single block or performed simultaneously, and/or the sequence of such blocks can be changed. For example, blocks **4100** and **4200** may be combined in some embodiments, such as where the face portion and at least one portion of the head body comprise a single piece of material. Block **4800** may be combined with one or more of blocks **4100**, **4300**, **4400**, **4500**, **4600**, and/or **4700** in the same or other examples, and may be achieved simultaneously by adjusting the center of gravity, the face height, the face size, the head volume, and/or one or more moments of inertia of the golf club head, such as via the mass redistribution mechanism of block **4800**. In the same or other examples, some of the blocks of method **4000** can be subdivided into several sub-blocks. For example, block **4100** can be subdivided into several sub-blocks for providing different portions of the head body of the golf club head. There can also be examples where method **4000** can comprise further or different blocks. As an example, method **4000** may comprise another block for providing or coupling a golf club shaft to the head body of block **4100**. In addition, there may be examples where method **4100** can comprise only part of the blocks described above. For example, one or more of blocks **4300**, **4400**, **4500**, **4600**, and/or **4700** may be optional in some implementations, and/or block **4800** may be skipped if not needed to achieve the requirements of block **4300**, block **4400**, block **4500**, block **4600**, and/or block **4700**. Other variations can be implemented for method **4000** without departing from the scope of the present disclosure.

(73) Although the golf club heads with optimized characteristics and related methods herein have been described with reference to specific embodiments, various changes may be made without departing from the spirit or scope of the present disclosure. For instance, while the above examples

may be described in connection with a driver-type golf club, the apparatus, methods, and articles of manufacture described herein may be applicable to other types of golf club such as a fairway wood-type golf club, a hybrid-type golf club, an iron-type golf club, a wedge-type golf club, or a putter-type golf club. Alternatively, the apparatus, methods, and articles of manufacture described herein may be applicable other type of sports equipment such as a hockey stick, a tennis racket, a fishing pole, a ski pole, etc.

(74) Additional examples of such changes and others have been given in the foregoing description. Other permutations of the different embodiments having one or more of the features of the various figures are likewise contemplated. Accordingly, the specification, claims, and drawings herein are intended to be illustrative of the scope of the disclosure and is not intended to be limiting. It is intended that the scope of this application shall be limited only to the extent required by the appended claims.

(75) The golf club heads with optimized characteristics and related methods discussed herein may be implemented in a variety of embodiments, and the foregoing discussion of certain of these embodiments does not necessarily represent a complete description of all possible embodiments. Rather, the detailed description of the drawings, and the drawings themselves, disclose at least one preferred embodiment, and may disclose alternative embodiments.

(76) Replacement of one or more claimed elements constitutes reconstruction and not repair. Additionally, benefits, other advantages, and solutions to problems have been described with regard to specific embodiments. The benefits, advantages, solutions to problems, and any element or elements that may cause any benefit, advantage, or solution to occur or become more pronounced, however, are not to be construed as critical, required, or essential features or elements of any or all of the claims, unless such benefits, advantages, solutions, or elements are expressly stated in such claims.

(77) As the rules to golf may change from time to time (e.g., new regulations may be adopted or old rules may be eliminated or modified by golf standard organizations and/or governing bodies such as the United States Golf Association (USGA), the Royal and Ancient Golf Club of St. Andrews (R&A), etc.), golf equipment related to the apparatus, methods, and articles of manufacture described herein may be conforming or non-conforming to the rules of golf at any particular time. Accordingly, golf equipment related to the apparatus, methods, and articles of manufacture described herein may be advertised, offered for sale, and/or sold as conforming or non-conforming golf equipment. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

(78) Moreover, embodiments and limitations disclosed herein are not dedicated to the public under the doctrine of dedication if the embodiments and/or limitations: (1) are not expressly claimed in the claims; and (2) are or are potentially equivalents of express elements and/or limitations in the claims under the doctrine of equivalents.

Claims

1. A golf club head comprising: a head body comprising: a head interior bounded by: a head front portion; a head rear portion; a head heel portion; a head toe portion; a head sole portion; and a head top portion; a hosel structure having a bore for receiving a golf club shaft, the bore having a hosel axis; a head volume magnitude; and an external weight structure located towards the head sole portion and the head rear portion of the head body, the external weight structure protruding at least partially from an external contour of the head sole portion of the head body and having a weight center; a face portion at the head front portion and comprising: a strikeface centerpoint; a strikeface perimeter; and a face height bounded by the strikeface perimeter, the strikeface centerpoint positioned at a midpoint of the face height; and a head center of gravity; wherein: when the golf club head is at an address position over a ground plane: a head vertical axis extends through the

head center of gravity and is orthogonal to the ground plane; a head horizontal axis extends through the head center of gravity, from the head heel portion to the head toe portion and is orthogonal to the head vertical axis; a loft plane of the golf club head is tangent to the strikeface centerpoint; a front plane of the golf club head extends through the strikeface centerpoint, parallel to the hosel axis, and orthogonal to the ground plane; a head depth plane extends through the strikeface centerpoint, parallel to the head horizontal axis and perpendicular to the loft plane; a CG height axis extends through the head center of gravity and intersects the head depth plane perpendicularly at a first intersection point; a CG height axis length is measured from an intersection of the head top portion exterior with the CG height axis to an intersection of the head sole portion exterior with the CG height axis; a head CG height of the head center of gravity is measured, along the CG height axis, between the head center of gravity and the first intersection point; a head CG depth of the head center of gravity is measured, parallel to the ground plane and orthogonal to the front plane, between: a second intersection point located at an intersection between the front plane and the ground plane; and a third intersection point located at an intersection between the head vertical axis and the ground plane; a hosel moment of inertia is measured about the hosel axis; a horizontal moment of inertia is measured about the head horizontal axis; a vertical moment of inertia measured about the head vertical axis; the head CG depth is approximately 35 mm to approximately 102 mm; the face height is approximately 40 mm to approximately 50 mm, measured parallel to the loft plane; and the golf club head comprises at least three performance characteristics selected from the group consisting of: a first performance characteristic, a second performance characteristic, a third performance characteristic, a fourth performance characteristic, and a fifth performance characteristic; wherein: the first performance characteristic comprises: the head CG height being less than or equal to approximately 5.08 mm; the second performance characteristic comprises: a CG performance ratio less than or equal to 0.56, as defined by (a) 76.2 mm minus the face height, divided by (b) the head CG Depth; the third performance characteristic comprises: (a) the head volume magnitude added to (b) a ratio between the head CG depth divided by an absolute value of the head CG height; wherein: the third performance characteristic is greater than or equal to 425 cc; the fourth performance characteristic comprises: the horizontal moment of inertia being greater than or equal to 39% of the hosel moment of inertia; and the fifth performance characteristic comprises: the vertical moment of inertia being greater than or equal to 59% of the hosel moment of inertia; a weight center elevation axis extends between the weight center and the ground plane, being orthogonal to the ground plane when the golf club head is at the address position; a head CG elevation is measured, along the head vertical axis, between the head center of gravity and the ground plane; a weight center elevation is measured, along the weight center elevation axis, between the head center of gravity and the ground plane; the weight center elevation is less than approximately 12.5 mm; a weight center depth of the weight center is measured, parallel to the ground plane, between: the second intersection point at the intersection between the front plane and the ground plane; and a fourth intersection point at an intersection between the weight center elevation axis and the ground plane; a depth ratio, defined by the weight center depth over the head CG depth, is less than 2.54; and an elevation ratio, defined by the weight center elevation over the head CG elevation, is greater than 0.44.

2. The golf club head of claim 1, wherein: the horizontal moment of inertia is approximately 2800 g*cm.sup.2 to approximately 4300 g*cm.sup.2.
3. The golf club head of claim 1, wherein: the weight center depth is greater than approximately 99.7 mm.
4. The golf club head of claim 1, wherein: the head CG depth is approximately 44 mm to approximately 102 mm.
5. The golf club head of claim 1, wherein: the head volume magnitude greater than or equal to 420 cc.
6. The golf club head of claim 1, wherein the head body comprises: a total head weight of

approximately 185 grams to approximately 225 grams.

7. The golf club head of claim 1, wherein: the head CG height is approximately 0 mm to 3.18 mm.

8. The golf club head of claim 1, wherein: the hosel moment of inertia is between approximately 7000 g*cm.sup.2 and approximately 11000 g*cm.sup.2.

9. The golf club head of claim 1 wherein: the head center of gravity is located between the head depth plane and the head sole portion.

10. The golf club head of claim 1, wherein: the face height is approximately 40 mm to approximately 50 mm, measured parallel to the loft plane.

11. A golf club head comprising: a head body comprising: a head interior bounded by: a head front portion; a head rear portion; a head heel portion; a head toe portion; a head sole portion; and a head top portion; a hosel structure having a bore for receiving a golf club shaft, the bore having a hosel axis; a head volume magnitude; and an external weight structure located towards the head sole portion and the head rear portion of the head body, the external weight structure protruding at least partially from an external contour of the head sole portion of the head body and having a weight center; a face portion at the head front portion and comprising: a strikeface centerpoint; a strikeface perimeter; and a face height bounded by the strikeface perimeter, the strikeface centerpoint positioned at a midpoint of the face height; and a head center of gravity; wherein: when the golf club head is at an address position over a ground plane: a head vertical axis extends through the head center of gravity and is orthogonal to the ground plane; a head horizontal axis extends through the head center of gravity, from the head heel portion to the head toe portion and is orthogonal to the head vertical axis; a loft plane of the golf club head is tangent to the strikeface centerpoint; a front plane of the golf club head extends through the strikeface centerpoint, parallel to the hosel axis, and orthogonal to the ground plane; a head depth plane extends through the strikeface centerpoint, parallel to the head horizontal axis and perpendicular to the loft plane; a CG height axis extends through the head center of gravity and intersects the head depth plane perpendicularly at a first intersection point; a CG height axis length is measured from an intersection of the head top portion exterior with the CG height axis to an intersection of the head sole portion exterior with the CG height axis; a head CG height of the head center of gravity is measured, along the CG height axis, between the head center of gravity and the first intersection point; a head CG depth of the head center of gravity is measured, parallel to the ground plane and orthogonal to the front plane, between: a second intersection point located at an intersection between the front plane and the ground plane; and a third intersection point located at an intersection between the head vertical axis and the ground plane; a hosel moment of inertia is measured about the hosel axis; a horizontal moment of inertia is measured about the head horizontal axis; a vertical moment of inertia measured about the head vertical axis; the head CG depth is approximately 35 mm to approximately 102 mm; the face height is approximately 40 mm to approximately 50 mm, measured parallel to the loft plane; and the golf club head comprises at least three performance characteristics selected from the group consisting of: a first performance characteristic, a second performance characteristic, a third performance characteristic, a fourth performance characteristic, and a fifth performance characteristic; wherein: the first performance characteristic comprises: the head CG height being less than or equal to approximately 5.08 mm; the second performance characteristic comprises: a CG performance ratio less than or equal to 0.56, as defined by (a) 76.2 mm minus the face height, divided by (b) the head CG Depth; the third performance characteristic comprises: (a) the head volume magnitude added to (b) a ratio between the head CG depth divided by an absolute value of the head CG height; wherein: the third performance characteristic is greater than or equal to 425 cc; the fourth performance characteristic comprises: the horizontal moment of inertia being greater than or equal to 39% of the hosel moment of inertia; and the fifth performance characteristic comprises: the vertical moment of inertia being greater than or equal to 59% of the hosel moment of inertia-; a weight center elevation axis extends between the weight center and the ground plane, being orthogonal to the ground plane when the golf club head is at the

address position; a head CG elevation is measured, along the head vertical axis, between the head center of gravity and the ground plane; a weight center elevation is measured, along the weight center elevation axis, between the head center of gravity and the ground plane; a weight center depth of the weight center is measured, parallel to the ground plane, between: the second intersection point at the intersection between the front plane and the ground plane; a fourth intersection point at an intersection between the weight center elevation axis and the ground plane; and wherein the weight center depth is greater than approximately 99.7 mm; a depth ratio, defined by the weight center depth over the head CG depth, is less than 2.54; and an elevation ratio, defined by the weight center elevation over the head CG elevation, is greater than 0.44.

12. The golf club head of claim 11 wherein: The head CG elevation is less than 28.5 mm.

13. The golf club head of claim 11 wherein: the hosel moment of inertia is between 7000 g*cm.sup.2 and 11,00 g*cm.sup.2.

14. The golf club head of claim 11 wherein: the external weight structure comprises a mass of approximately 2 grams to 50 grams; and the external weight structure comprises a volume of approximately 1 cc to 30 cc.

15. The golf club head of claim 11, wherein: a clock grid comprises at least: a 12 o'clock ray; a 3 o'clock ray; a 4 o'clock ray; a 5 o'clock ray; an 8 o'clock ray; and a 9 o'clock ray; when the golf club head is at the address position over the ground plane, from a bottom view of the golf club head, the 12 o'clock ray is aligned with the strikeface centerpoint and orthogonal to a front intersection line between the loft plane and the ground plane; the clock grid is centered along the 12 o'clock ray, at a midpoint between a front end of the head front portion and a rear end of the head rear portion; the 3 o'clock ray extends towards the head heel portion; the 9 o'clock ray extends towards the head toe portion; a perimeter of the external weight structure is at least partially bounded by the 4 o'clock ray and the 9 o'clock ray; and a center of gravity of the external weight structure is located between the 5 o'clock ray and the 8 o'clock ray.
